ECEN 123/CSEN 123/MECH 143 – Mechatronics Lab

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List of components:
Arduino
8x LEDs (Not specified which, but we're going to use this one)
LED DIP Pack
DIP Switch
4x (100k) Resistor
ResistorSIP

Identify 8 pins on the Arduino board that can be used as digital outputs and one pin that can be used as a digital input.
Pins 2-9 can be used as digital output pins
Pin 10 can be used as a digital input
(all digital pins are inputs and outputs)

Calculate the typical current flowing through the LED in the circuit on the next page when the LED is on.
Voltage input: 5V
Forward Voltage Drop: 1.8V at 2mA
Resistor: 330ohms
Expected current: (5-1.8)/330 = 9.70mA
Because expected current is greater than 2mA, we expect the forward voltage to actually be a little greater than 1.8V (based on the IV curve) which would then lead to a slightly lower current. Using the IV curve in the datasheet, it looks like we actually are likely to see a current closer to 9.4mA with a Vf of 1.9V.

We used the lite-on red LED for this calculation, and if we use a different one in lab, the expected current will be different due to different Vfs. They're all around 2V though, so not a huge deviation.

Calculate the typical current flowing through the LED in the circuit on the next page when the LED is off.
The current will be 0. The diode (LED) won't let any current pass through unless it's on.

Review the documentation for the 11-pin SIP resistor. Explain how pin 1 is different from the other pins.

Pin 1 ties all of the resistors together, so this pin can be grounded to complete the circuit.

Take a selfie of you and your lab partners having your lab planning meeting (in person). Include the photo in your prelab report.